POINCARE DUALITY FOR THE DE RHAM COHOMOLOGY OF A SMOOTH PROPER EQUIDIMENSIONAL SCHEME OVER A FIELD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file derham.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-derham/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The Hodge-to-de Rham spectral sequence. Same chapter derham.tex, Section \ref{section-hodge-to-de-rham} (invoked at proof line 4709). It relates the de Rham cohomology of the smooth proper scheme to the cohomology of the sheaves of differentials, which is how the extreme degrees are identified.
- Vanishing of the cohomology of a coherent sheaf in degrees above the dimension, and the degeneration of the extreme parts of the spectral sequence. Cohomology, label cohomology-proposition-vanishing-Noetherian (invoked at proof line 4711) and Same chapter derham.tex, labels lemma-bottom-part-degenerates and lemma-top-part-degenerates (invoked at proof lines 4712 and 4713). Together with the vanishing of Ω^i for $i > d$ they give $H^0_{dR}(X/k) = H^0(X, \mathcal{O}_X)$ and $H^d(X, \Omega^d_{X/k}) = H^{2d}_{dR}(X/k)$.
- The Hodge duality pairing in the extreme case. Same chapter derham.tex, label lemma-duality-hodge (invoked at proof lines 4724 and 4768). It supplies the trace map in the base case $p = 0$ of the induction, giving perfect pairings between $H^i(X, \mathcal{O}_X)$ and $H^{d-i}(X, \Omega^d)$.
- The wedge product on the total complex and the construction of cup products. Same chapter derham.tex, equation equation-wedge (invoked at proof line 4731) and Section \ref{section-cup-product} (invoked at proof line 4752). The wedge map on $\operatorname{Tot}(\Omega^\bullet \otimes \Omega^\bullet)$ restricts to the maps γ_p whose induced products are the pairings studied in the induction.

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Source reference. Original proposition 02UZ: sources/stacks-chow/tags.txt, source lines 3750–3750. Navigation pointer for \ref{cohomology-proposition-vanishing-Noetherian} used inside the counted span (line 4711); the argument that consumes it is retained in the span.

Source reference. Original equation 0FM2: sources/stacks-derham/pinned-original-source.tex, source lines 495–495. Navigation pointer for \ref{equation-wedge} used inside the counted span (line 4731); the argument that consumes it is retained in the span.

Source reference. Original lemma 0FW5: sources/stacks-derham/pinned-original-source.tex, source lines 4548–4548. Navigation pointer for \ref{lemma-bottom-part-degenerates} used inside the counted span (line 4712); the argument that consumes it is retained in the span.

Source reference. Original lemma 0FW4: sources/stacks-derham/pinned-original-source.tex, source lines 4496–4496. Navigation pointer for \ref{lemma-duality-hodge} used inside the counted span (line 4724); the argument that consumes it is retained in the span.

Source reference. Original lemma 0FW6: sources/stacks-derham/pinned-original-source.tex, source lines 4577–4577. Navigation pointer for \ref{lemma-top-part-degenerates} used inside the counted span (line 4713); the argument that consumes it is retained in the span.

Source reference. Original section 0FM1: sources/stacks-derham/pinned-original-source.tex, source lines 483–483. Navigation pointer for \ref{section-cup-product} used inside the counted span (line 4752); the argument that consumes it is retained in the span.

Source reference. Original section 0FM6: sources/stacks-derham/pinned-original-source.tex, source lines 671–671. Navigation pointer for \ref{section-hodge-to-de-rham} used inside the counted span (line 4709); the argument that consumes it is retained in the span.

Proposition 0.1. *Let k be a field. Let X be a nonempty smooth proper scheme over k equidimensional of dimension d. There exists a k-linear map*

$$t : H^{2d}_{dR}(X/k) \longrightarrow k$$

unique up to precomposing by multiplication by a unit of $H^0(X, \mathcal{O}_X)$ with the following property: for all i the pairing

$$H^i_{dR}(X/k) \times H^{2d-i}_{dR}(X/k) \longrightarrow k, \quad (\xi, \xi') \longmapsto t(\xi \cup \xi')$$

is perfect.

Proof. By the Hodge-to-de Rham spectral sequence (Section 0FM6), the vanishing of $\Omega^i_{X/k}$ for $i > d$, the vanishing in Cohomology, Proposition 02UZ and the results of Lemmas 0FW5 and 0FW6 we see that $H^0_{dR}(X/k) = H^0(X, \mathcal{O}_X)$ and $H^d(X, \Omega^d_{X/k}) = H^{2d}_{dR}(X/k)$. More precisely, these identifications come from the maps of complexes

$$\Omega^\bullet_{X/k} \to \mathcal{O}_X[0] \quad \text{and} \quad \Omega^d_{X/k}[-d] \to \Omega^\bullet_{X/k}$$

Let us choose $t : H^{2d}_{dR}(X/k) \to k$ which via this identification corresponds to a t as in Lemma 0FW4. Then in any case we see that the pairing displayed in the lemma is perfect for $i = 0$.

Denote $\underline{k}$ the constant sheaf with value k on X. Let us abbreviate $\Omega^\bullet = \Omega^\bullet_{X/k}$. Consider the map (0FM2) which in our situation reads

$$\wedge : \mathrm{Tot}(\Omega^\bullet \otimes_{\underline{k}} \Omega^\bullet) \longrightarrow \Omega^\bullet$$

For every integer $p = 0, 1, \ldots, d$ this map annihilates the subcomplex $\mathrm{Tot}(\sigma_{>p}\Omega^\bullet \otimes_{\underline{k}} \sigma_{\geq d-p}\Omega^\bullet)$ for degree reasons. Hence we find that the restriction of $\wedge$ to the subcomplex $\mathrm{Tot}(\Omega^\bullet \otimes_{\underline{k}} \sigma_{\geq d-p}\Omega^\bullet)$ factors through a map of complexes

$$\gamma_p : \mathrm{Tot}(\sigma_{\leq p}\Omega^\bullet \otimes_{\underline{k}} \sigma_{\geq d-p}\Omega^\bullet) \longrightarrow \Omega^\bullet$$

Using the same procedure as in Section 0FM1 we obtain cup products

$$H^i(X, \sigma_{\leq p}\Omega^\bullet) \times H^{2d-i}(X, \sigma_{\geq d-p}\Omega^\bullet) \longrightarrow H^{2d}_{dR}(X, \Omega^\bullet)$$

We will prove by induction on p that these cup products via t induce perfect pairings between $H^i(X, \sigma_{\leq p}\Omega^\bullet)$ and $H^{2d-i}(X, \sigma_{\geq d-p}\Omega^\bullet)$. For $p = d$ this is the assertion of the proposition.

The base case is $p = 0$. In this case we simply obtain the pairing between $H^i(X, \mathcal{O}_X)$ and $H^{d-i}(X, \Omega^d)$ of Lemma 0FW4 and the result is true.

Induction step. Say we know the result is true for p. Then we consider the distinguished triangle

$$\Omega^{p+1}[-p-1] \to \sigma_{\leq p+1}\Omega^\bullet \to \sigma_{\leq p}\Omega^\bullet \to \Omega^{p+1}[-p]$$

and the distinguished triangle

$$\sigma_{\geq d-p}\Omega^\bullet \to \sigma_{\geq d-p-1}\Omega^\bullet \to \Omega^{d-p-1}[-d+p+1] \to (\sigma_{\geq d-p}\Omega^\bullet)[1]$$

Observe that both are distinguished triangles in the homotopy category of complexes of sheaves of $\underline{k}$-modules; in particular the maps $\sigma_{\leq p}\Omega^\bullet \to \Omega^{p+1}[-p]$ and $\Omega^{d-p-1}[-d+p+1] \to (\sigma_{\geq d-p}\Omega^\bullet)[1]$ are given by actual maps of complexes, namely using the differential $\Omega^p \to \Omega^{p+1}$ and the differential $\Omega^{d-p-1} \to \Omega^{d-p}$. Consider the long exact cohomology sequences associated to these distinguished triangles

$$\begin{array}{ccc}
H^{i-1}(X, \sigma_{\leq p}\Omega^\bullet) & & H^{2d-i+1}(X, \sigma_{\geq d-p}\Omega^\bullet) \\
\downarrow{\scriptstyle a} & & \uparrow{\scriptstyle a'} \\
H^i(X, \Omega^{p+1}[-p-1]) & & H^{2d-i}(X, \Omega^{d-p-1}[-d+p+1]) \\
\downarrow{\scriptstyle b} & & \uparrow{\scriptstyle b'} \\
H^i(X, \sigma_{\leq p+1}\Omega^\bullet) & & H^{2d-i}(X, \sigma_{\geq d-p-1}\Omega^\bullet) \\
\downarrow{\scriptstyle c} & & \uparrow{\scriptstyle c'} \\
H^i(X, \sigma_{\leq p}\Omega^\bullet) & & H^{2d-i}(X, \sigma_{\geq d-p}\Omega^\bullet) \\
\downarrow{\scriptstyle d} & & \uparrow{\scriptstyle d'} \\
H^{i+1}(X, \Omega^{p+1}[-p-1]) & & H^{2d-i-1}(X, \Omega^{d-p-1}[-d+p+1])
\end{array}$$

By induction and Lemma 0FW4 we know that the pairings constructed above between the k-vectorspaces on the first, second, fourth, and fifth rows are perfect. By the 5-lemma, in order to show that the pairing between the cohomology groups in the middle row is perfect, it suffices to show that the pairs (a, a'), (b, b'), (c, c'), and (d, d') are compatible with the given pairings (see below).

Let us prove this for the pair (c, c'). Here we observe simply that we have a commutative diagram

$$\begin{array}{ccc} \mathrm{Tot}(\sigma_{\leq p}\Omega^\bullet \otimes_{\underline{k}} \sigma_{\geq d-p}\Omega^\bullet) & \longleftarrow & \mathrm{Tot}(\sigma_{\leq p+1}\Omega^\bullet \otimes_{\underline{k}} \sigma_{\geq d-p}\Omega^\bullet) \\ \downarrow{\scriptstyle \gamma_p} & & \downarrow \\ \Omega^\bullet & \xleftarrow{\gamma_{p+1}} & \mathrm{Tot}(\sigma_{\leq p+1}\Omega^\bullet \otimes_{\underline{k}} \sigma_{\geq d-p-1}\Omega^\bullet) \end{array}$$

Hence if we have $\alpha \in H^i(X, \sigma_{\leq p+1}\Omega^\bullet)$ and $\beta \in H^{2d-i}(X, \sigma_{\geq d-p}\Omega^\bullet)$ then we get $\gamma_p(\alpha \cup c'(\beta)) = \gamma_{p+1}(c(\alpha) \cup \beta)$ by functoriality of the cup product.

Similarly for the pair (b, b') we use the commutative diagram

$$\begin{array}{ccc} \mathrm{Tot}(\sigma_{\leq p+1}\Omega^\bullet \otimes_{\underline{k}} \sigma_{\geq d-p-1}\Omega^\bullet) & \longleftarrow & \mathrm{Tot}(\Omega^{p+1}[-p-1] \otimes_{\underline{k}} \sigma_{\geq d-p-1}\Omega^\bullet) \\ \downarrow{\scriptstyle \gamma_{p+1}} & & \downarrow \\ \Omega^\bullet & \xleftarrow{\wedge} & \Omega^{p+1}[-p-1] \otimes_{\underline{k}} \Omega^{d-p-1}[-d+p+1] \end{array}$$

and argue in the same manner.

For the pair (d, d') we use the commutative diagram

$$\begin{array}{ccc} \Omega^{p+1}[-p] \otimes_{\underline{k}} \Omega^{d-p-1}[-d+p] & \longleftarrow & \mathrm{Tot}(\sigma_{\leq p}\Omega^\bullet \otimes_{\underline{k}} \Omega^{d-p-1}[-d+p]) \\ \downarrow & & \downarrow \\ \Omega^\bullet & \longleftarrow & \mathrm{Tot}(\sigma_{\leq p}\Omega^\bullet \otimes_{\underline{k}} \sigma_{\geq d-p}\Omega^\bullet) \end{array}$$

and we look at cohomology classes in $H^i(X, \sigma_{\leq p}\Omega^\bullet)$ and $H^{2d-i}(X, \Omega^{d-p-1}[-d+p])$. Changing i to $i-1$ we get the result for the pair (a, a') thereby finishing the proof that our pairings are perfect.

We omit the argument showing the uniqueness of t up to precomposing by multiplication by a unit in $H^0(X, \mathcal{O}_X)$. $\square$